Template Method

Mine CSV/JSON reports with shared pipeline

Season 3 · GoF · Beginner

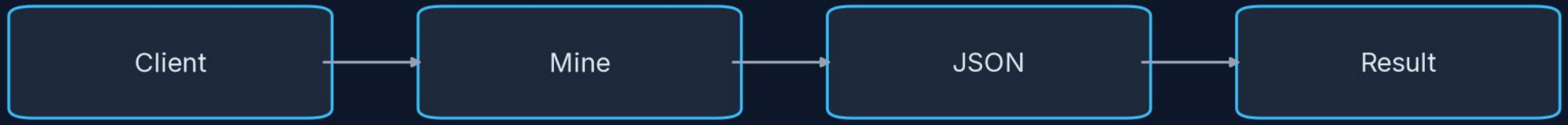
Today's Hook

- Problem: Mine CSV/JSON reports with shared pipeline
- Naive pain cue: Failure path is asserted in ProblemTest (injected fault / invalid input)
- We will refactor to Template Method with a runnable Demo.
- Failure path is locked by ProblemTest.

The Problem

- Scenario: Mine CSV/JSON reports with shared pipeline
- Constraints: solve «Mine CSV/JSON reports with shared pipeline» with Template Method — keep the demo runnable and the failure path observable.
- Naive pain: Naive version fails to cleanly solve: Mine CSV/JSON reports with shared pipeline
- Success looks like: rows=2

Structure



Template Method: Mine CSV/JSON reports with shared pipeline

Real-World Scenario

- Running example: Mine CSV/JSON reports with shared pipeline
- Key types: DataMiner, CsvMiner, JsonMiner
- Happy-path stdout cue: rows=2
- Failure-path stdout cue: Failure path is asserted in ProblemTest (injected fault / invalid input)

Naive Code

```java
package com.buildwithwaqar.patterns.gof.behavioral.templatemethod.problem;

import java.util.*;

public final class NaiveMine {
  public static String csv(String raw){ return "rows=" + raw.split("\n").length; }
  public static String json(String raw){ return "rows=" + raw.split("\\{", -1).length; }
}
```

Why Naive Hurts

- Naive version fails to cleanly solve: Mine CSV/JSON reports with shared pipeline
- Without Template Method, call sites absorb the wrong details.
- Failure mode to show on camera: Failure path is asserted in ProblemTest (injected fault / invalid input)
- New requirements force shotgun edits across the naive code.

Solution

```
package com.buildwithwaqar.patterns.gof.behavioral.templatemethod.problem;

import java.util.*;
import java.util.concurrent.*;
import java.util.concurrent.atomic.*;
import java.util.function.*;

abstract class DataMiner {
  final String mine(String raw){ String opened=open(raw); List<String> rows=parse(opened); return analyze(rows); }
  abstract String open(String raw); abstract List<String> parse(String opened);
  String analyze(List<String> rows){ return "rows="+rows.size(); }
}
final class CsvMiner extends DataMiner {
  String open(String raw){ return raw; }
  List<String> parse(String opened){ return Arrays.stream(opened.split("\n")).filter(s->!s.isBlank()).toList(); }
}
final class JsonMiner extends DataMiner {
  String open(String raw){ return raw; }
  List<String> parse(String opened){ return Arrays.stream(opened.split("\n")).filter(s->s.contains(":")).toList(); }
}
```

Runnable main

```
package com.buildwithwaqar.patterns.gof.behavioral.templatemethod.problem;
import java.util.*; import java.util.concurrent.*;
public final class ProblemDemo {
  public static void main(String[] args) throws Exception {
    System.out.println("=== Template Method: CSV/JSON mine ===");
    System.out.println("--- naive ---"); System.out.println(NaiveMine.csv("a\nb"));
System.out.println("--- template ---");
System.out.println(new CsvMiner().mine("a\nb\n"));
System.out.println(new JsonMiner().mine("{\n x:1\n}\n"));
  }
}
```

Walkthrough Output

```
=== Template Method: CSV/JSON mine ===
--- naive ---
rows=2
--- template ---
rows=2
rows=1
```

Step-by-Step Trace

- Match stdout to stages — look for: rows=2
- Then the injected failure: Failure path is asserted in ProblemTest (injected fault / invalid input)
- Re-run: java -cp ... 24-template-method ProblemDemo (see episode folder).
- Pause the video and predict the next printed line.

Modern Java 25 / Zazr

```
// Java 25 direction for Template Method
// Prefer records + sealed where they shrink the demo.
// Problem: Mine CSV/JSON reports with shared pipeline
```

Trade-offs

- Use Template Method when «Mine CSV/JSON reports with shared pipeline» repeats and a simpler feature does not erase it.
- Refuse extra types when a sealed switch or a function value already suffices.
- Teaching demos omit production metrics, retries, and multi-region clocks.
- Interview soundbite: state the force behind Template Method in product language.

Related Patterns

- Name the pattern usually confused with Template Method and separate the forces.
- Sketch the roles for «Mine CSV/JSON reports with shared pipeline» from memory in sixty seconds.
- Point to the next episode in this season on the playlist map.
- Link this episode's series/ folder in the YouTube description.

Recap

- Problem: Mine CSV/JSON reports with shared pipeline
- Naive pain → Template Method boundary → Demo stdout.
- Remember the failure cue: Failure path is asserted in ProblemTest (injected fault / invalid input)
- Break a ProblemTest assertion, watch it fail, restore it.